

**SST BASED SMART HELMET FOR
CYCLISTS AND MOTOR CYCLISTS: A CASE STUDY**

25-26J-294

Project Proposal Report

Samarakoon S.S.A.D.S.B

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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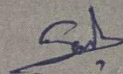
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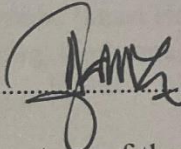
August 2025

Declaration of The Candidate & Supervisor

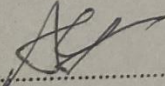
I declare that this is my work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


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29/08/2025
Date


Signature of the co-supervisor

29/08/25
Date

Acknowledgment

My research supervisor, Mr. Jagath Wickramarathne, and my co-supervisor, Mr. Amila Nuwan Alexander, provided invaluable support that made this publication and the research behind it feasible. They have inspired me and helped me stay on task with their passion, expertise, and meticulous attention to detail. I want to express my gratitude to my supervisors as well as the other members of my group, Vithanage K.W.Y.L.N., Samarasinghe K.P.C., and Primasha W.G.R., for their support, encouragement, and enlightening advice. Lastly, a special thanks to my parents, who never fail to provide love and support.

Abstract

Motor riders, especially those who perform long-distance commutes, those in professional courier business and those who encounter much traffic in commuting, often find themselves in stressful riding conditions that may affect their cognitive ability, delay their response time and make them prone to accidents. This study is devoted to the development of a stress monitoring system based on electroencephalography that can be embedded into a smart helmet to help detect when a rider is in a hazardous mental state and take a proactive approach to intervene hence helping to improve the rider safety. Each time, the system reads the brainwave activity by means of a small electroencephalogram (EEG) brainwave sensor module built into the helmet. Transmission and reception of data is in real time to a companion mobile application via Bluetooth Low Energy (BLE) which was used to arrive at efficient and defective free communication. Using EEG patterns, a machine learning model trained to identify high stress or emotional strain based on patterns specific to each rider identifies the levels of observable stress. When it is detected, the system will alert instantly, and may offer personally tailored suggestions, including guided breathing, switching to calming music or ambient sounds, or suggesting a safe stop for rest. The mobile application provides live information with color-coded alerts, on the emotional state of the rider and stress level alerts so that one will always know, as it facilitates informed decision making. The targeted audience who will use this type of helmet are the delivery riders working in a professional sector, police motor units, first responders, and ordinary people who have to use helmets in high-risk traffic environments. With the use of physiological data capture and analysis, AI and fluid mobile connectivity, this system will revise the role of helmets as a passive mechanism of injury prevention and make them active entities that support mental well-being. The given solution is a meaningful leap in the development of personal safety equipment, facilitating both the prevention of accidents and the overall cognitive state of riders as one of the long-term goals.

Keywords: Smart Helmet, EEG-based stress monitoring, Bluetooth Low Energy (BLE), Machine Learning, Real-time stress detection, Rider safety, Accident Prevention.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
EEG	Electroencephalogram
ESP32	Espressif Systems 32-bit Microcontroller
GSM	Global System for Mobile Communications
GPS	Global Positioning System
IDE	Integrated Development Environment
IoT	Internet of Thing
ML	Machine Learning
WHO	World Health Organization

1. INTRODUCTION

1.1. Background & Literature Survey

The safety of motorcycles has primarily been dependent on the passive measures of physical protection like helmets and riding apparel that do not prevent getting into an accident per se but tend to minimize the extent of injury. These approaches are very useful, but they do not help in cases where the rider is not displaying the signs of being under pressure, tired or experiencing cognitive overload, which may hinder their judgment, affect their reaction time as well as decrease their level of awareness. Work on wearable biosensing technology in the past few years has led to real-time, non-invasive measures of activity in the brain, including electroencephalography (EEG). Together with Bluetooth Low Energy (BLE), these systems can send physiology measurements to mobile applications in real time to be processed and visualized, allowing feedback to be provided in real time without distracting the rider.

The literature in the area demonstrates an increasing agreement on the necessity of real-time monitoring of the mental state when it comes to road safety and the well-being of the riders. Studies using EEG reveal that machine learning models are able to recognize the pattern of stress and intervene with guiding breathing or relaxation prompts, or safe-stop suggestions. Studies in automotive and occupational safety have emphasized that stress prevention is the best solution to curb accident occurrences, boost decision making and increase endurance. These results indicate the possible inclusion of EEG-based stress monitoring in smart helmets to transition them to active protective smart systems that will ensure the safety of not only the physical but also the cognitive health of riders.

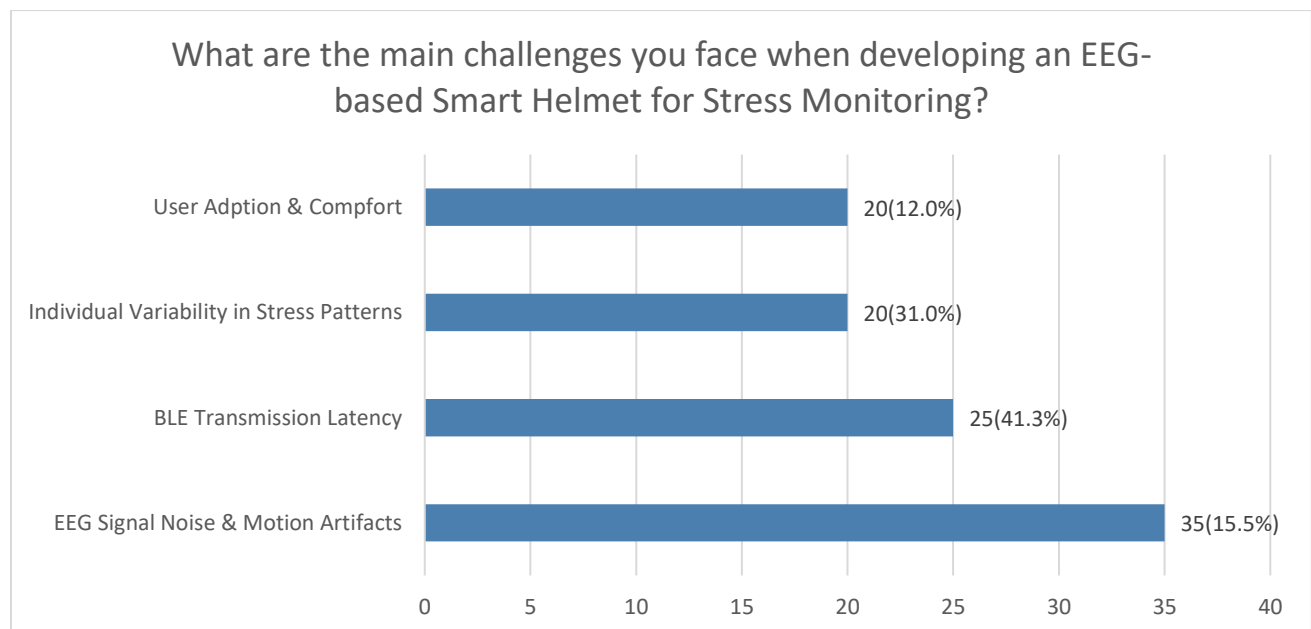


Figure 1: Summary of response challenges faced when monitoring stress.

Though there is strong potential in EEG-based monitoring to ensure the safety of riders, there are numerous technical and even practical challenges to developing a smart helmet. EEG signal noise and motion artifacts (35%) are the most significant probable reasons as depicted in the chart as resulted due to the head motions, vibrations, and environmental influence when riding a motorcycle (3). Such distortions decrease the dependability of stress class and underline the importance of advanced signal processing procedures to guarantee integrity of information.

Latency of Bluetooth Low Energy (BLE) transmission (25%) is another key issue since time delays or packet losses may interfere with real-time feedback. As continuous monitoring is based on the smooth communication between the helmet and the mobile application, it is necessary to optimize home transmission protocols to maximize the reduction of disruptions. Also, a matter of high importance is the issue of person-to-person differences in stress patterns (20%). EEG signals vary with body riders, so it becomes ineffective to use universal machine learning-based models, making the use of adaptive and individual learning methods necessary.

Last was the factor of user adoption and comfort (20%) which would carry a significant challenge to mainstream use. Riders will be reluctant to wear a helmet which is heavy, intrusive or not very pretty although it may include advanced features. The design must be thoughtful considering this challenge and providing evidence to prove its benefits like proactive stress warnings and well-being, among others. Collectively, these obstacles depict the difficulty of adopting EEG-driven smart helmets and, at the same time, point to the potential of brand-new solutions that couple AI and biosensing with mobile connectivity to revolutionize motorcycle safety.

1.2. Literature Survey

Rider safety and cognitive state monitoring are the areas that the published body of literature has advanced beyond being protective (helmet, clothing) to the active (sensors, system) stage that enables risk detection and mitigation prior to a crash. The use of wearable electroencephalography (EEG) is a viable approach of quantifying stress and workload on a real-time basis; EEG-stress outcomes have been agreed upon in challenging tasks, representative of a real-life atmosphere, and on-road driving conditions, indicating its applicability under such conditions [1].

In recent reviews and empirical studies, EEG-based stress and emotion recognition has been demonstrated to be able to reach high temporal resolution and make use of machine-learning pipelines to suit nonstationary neural signals. They describe recent advances in feature engineering, deep learning and cross-subject generalization and underscore ongoing issues with robustness beyond the laboratory [2], [3]. Meanwhile, there have been developments in the field of smart helmets, where the feasibility of placing biopotential electrodes in helmet liners to detect neural and cardiac functions is shown to be possible- indication that the form factor needed to become a product in the motorcycle space might not involve intrusive systems. Scoping reviews also outline an emerging trend of multimodal helmets integrating sensing, analytics and connectivity health and safety [4], [5].

One of the technical milestones to overcome in-motion EEG is motion and muscle artifact contamination due to movements of the head, vibration, and dynamics. Comparative and survey articles outline the requirement on artifact-extraction techniques (e.g., adaptive filtering, ICA, reference sensors, and contemporary flavoring methods) to protect nerve cells through the mobile surroundings [6], [7]. Stable wireless telemetry is also crucial. Surveys of Bluetooth Low Energy (BLE) in remote health monitoring and special-purpose, EEG-specific, architectures report low-latency mobile streaming and low-power operation, and describe design decisions (e.g. on packetization, buffering, re-connection) to enable continuous analytics on devices [8], [9].

On the modeling side, the literature is unanimous that there is high inter-individual variability in the EEG patterns, and, due to that, there has been effort to develop personalized models or domain-adaptation techniques to keep the domain of riders and sessions. New cross-subject and cross-dataset techniques (e.g., adaptation networks and spatial feature learning) have better generalization, a requirement in the domain [10], [11], since rider-particular calibration and lifelong learning are necessary. In addition to detection, in-ride interventions are important: human-factors studies warn on feedback intrusiveness and short feedback duration, and driver-stress studies show that guided breathing feedback (haptic/voice) can relieve stress without impairing driving performance-an intervention cadence that ideally suits motorcyclists through subliminal haptics or aural, abridged notices [12], [13].

Collectively, the literature suggests that the incorporation of helmet-embedded EEGs, low-latency streaming features using BLE, stress characterization and classification that is user-specific, and an intervention that is distractive is novel and necessary particularly in motorcyclists. Though conceptually possible to sense biopotentials via helmet as evidenced by proofs-of-concept and manifold evidence attesting to the efficacy of EEG-derived stress detection and intervention in driving, a common linkage, a rider-centric approach that with rider data and feedback, could close a gap in research by linking the data and intermediary to the end action on ride, is also a visible field of research your component is likely to bridge [4], [1], [5].

1.3. Research Gap

In this study, we attempt to shed some light on an important knowledge gap in the field of rider safety, which resides in the relative lack of incorporation of EEG stress detection techniques into smart helmets that can be used on a real-time basis. Although other researchers confirmed EEG as being effective in stress detection and measured the helmet sensors independently, little is known about the combined use of EEG and stress sensing in real-world shielding environments to provide dependable detection, concurrent wireless transmission, and applicable, non-invasive interventions. There is a critical need to fill this gap in the development of both the academic and the industry-practical contributions to proactive motorcyclist safety innovations.

Research Paper	EEG Stress Detection Accuracy	Motion Artifact Removal	BLE Data Transmission	Personalized Stress Models	Non-Distracting Interventions
[1] Sciaraffa, N., et al. (2022). Validation of a Light EEG-Based Measure for Real-Time Stress in Driving.	Yes	No	No	No	No
[2] Katmah, R., et al. (2021). A Review on Mental Stress Assessment Methods Using EEG.	Yes	No	No	No	No
[3] Gkintoni, E., et al. (2025). A Systematic Review of EEG-Based Emotion Recognition.	Yes	No	No	Yes	No
[4] Von Rosenberg, W., et al. (2016). Smart Helmet: Wearable Multichannel ECG and EEG.	Yes	No	No	No	No
[5] Lee, P., & Kim, H. (2022). Trends in Smart Helmets with Multimodal Sensing for Health and Safety.	Yes	No	Yes	No	No
[6] Seok, D., et al. (2021). Motion Artifact Removal Techniques for Wearable EEG and PPG.	No	Yes	No	No	No

[7] Jiang, X., et al. (2019). Removal of Artifacts from EEG Signals: A Review.	No	Yes	No	No	No
[8] Fourati, L. C., et al. (2020). Remote Health Monitoring Systems Based on BLE: A Survey.	No	No	Yes	No	No
[9] EEG-over-BLE: Low-Latency, Low-Power Architecture for Multi-Channel EEG (2022–2023).	No	No	Yes	No	No
[10] Lu, W., et al. (2024). Domain-Adaptation Spatial Feature Perception Network for Cross-Subject EEG.	Yes	No	No	Yes	No
[11] Imtiaz, M. N., et al. (2025). Enhanced Cross-Dataset EEG- Based Emotion Recognition.	Yes	No	No	Yes	No
[12] Paredes, P. E., et al. (2018). Just Breathe: In-Car Interventions for Guided Slow Breathing.	No	No	No	No	Yes
[13] Balters, S., et al. (2020). Calm Commute: Guided Slow Breathing for Drivers.	No	No	No	No	Yes
Proposed System (EEG-Based Smart Helmet for Real-Time Stress Monitoring)	Yes	Yes	Yes	Yes	Yes

Table 1: Comparison with existing system

In this table:

- "No" indicates that the paper does not address the specified challenge.
- "Yes" indicates that the paper addresses the specified challenge.

1.4. Research Problem

Riders are always subjected to a severe stress situation, including numerous traffic jams, long journeys and a stressful working environment. These stress factors adversely affect thinking, response times and awareness thus predisposing accidents. There are traditional safety protection methods, which are helmets and protective clothing, and they are a type of passive protection that can only mitigate and alleviate the impact of an accident. They fail to consider the psychological condition of the rider which can determine the outcome of mostly preventing an accident. This makes new safety solutions that go beyond physical protection necessary and involve proactive sensing of cognitive well-being.

The proposed research is expected to fill this gap by examining screening of the motorcyclist in real-time regarding his mental and emotional stress condition with the help of EEG-based wearable technology in the form of a smart helmet. The proposed system will constantly monitor the brainwave, predict patterns of stress and give timely intervention strategies that will reduce the chances of getting into accidents especially. Nonetheless, this is a very complicated task since a few severe difficulties are involved. Due to the high exposure of the EEG signals to motion artifacts and environmental noise caused when the motorcyclist is riding, their sensitivity becomes the first issue. This may interfere with the accuracy of the classification because it is hard to delink accurate stress responses and that of distortions brought by vibrations or head movement, and external stimuli. Also, the existence of great variability in physiological and behavioral reactions of the riders makes it even harder to come up with a uniform model. Stress affects various people in various ways thus adaptive approaches to this situation must be very personal otherwise significant accuracy might not be achieved at all.

Besides classification, the system should also have the potential to provide interventions in an immediate, dependable, and less obtrusive fashion. The alerts, relaxation signals, or safe-stop recommendations that must be offered in the companion mobile application should not cause delays or communication failures because the delay may diminish the results of stress control and introduce safety-related issues. As such, the research problem is highlighted in developing a viable, proficient, and user-oriented smart helmet device that can break through signal noise, subject variance, and interpersonal communication to present effective and on-time interventions. The solution to solving this is multidisciplinary, combining wearable EEG, machine learning, advanced signal processing, and human-computer interaction in order to converge two worlds- making helmets not only reactive, but also adaptive safety devices, protecting the physical and cognitive health of the people riding them.

2. OBJECTIVES

2.1 Main Objective

Introducing a smart helmet that has electroencephalography (EEG) technology, where the main aim consists in the promotion of safety and welfare of motorcyclists to be viewed as a progressive and essential step. The helmet has a carefully considered design to overcome these psychological and emotional barriers of long commuting, courier services, and unsafe traffic conditions. Its design incorporates novel biosensors, wireless communication units and intelligent programming to monitor and intercept the dangers posed by lapses caused by stress in the rider awareness and responsiveness.

Central to this is the fact that the smart helmet will be fitted with a lightweight EEG that will strategically be embedded in the helmet padding to similarly record the brainwave signals constantly during the rides. These sensors are read in real-time capturing and analyzing the pattern of mental activity which has a correlation to increased stress or emotional pressure. Once it is detected, the system quickly analyzes this data and wirelessly via Bluetooth Low Energy (BLE) sends it to a companion mobile application, where there is an efficient transfer of data with no disruption or loss.

The problem of classification does not arise, as the helmet utilizes a trained machine learning model which can tell the difference between a normal cognitive state and an elevated one of stress. The model therefore adjusts to the individual variability of each rider by analyzing the EEG pattern, increasing the accuracy on detection and reducing false positives or negatives. As soon as excessive levels of stress are detected, the system switches on those interventions meant to increase safety and comfort. These may include indirect cues and breathing instructions or customized recommendations like getting a break, drinking water, changing the position or listening to relaxing sounds.

In the same manner, the smart helmet is an important safety precaution in professional and high-risk riding settings such as in police departments or delivery riders or first responders where constant cognitive strain is present. Leveraging its round-the-clock monitoring scenario, it has the chance to keep riders conscious of the state of mind they are in and notify them to do something that lowers stress levels when the time comes. Moreover, the mobile application is developed to deliver live information, the color-marked indicators of the level of stress that will allow riders to improve their decisions during the ride. Updates that are provided on the course post-ride also enable one to reflect on stress levels, which are crucial to wellness in the long run.

Also, the combination of EEG technology with BLE communication highlights the possibility of this system to make the traditional helmets not only passive protective equipment but also active safety equipment. The smart helmet goes beyond becoming an accident-preventing device in that, with the assistance of wearable sensing, artificial intelligence, and attention to human-centered design, it also aims to enhance cognitive well-being among riders.

In brief, the development of this EEG-integrated smart helmet forms the proactive approach to the safety of motorcyclists. The system will seek to safeguard both the physical and mental health of riders by incorporating constant biosensing and proactive task-related analysis along with mobile phone accessibility in order that the system becomes a vital type of protection in visiting and returning to work, developing safer and healthier trips.

2.2 Specific Objectives

To achieve the main objective, the specific objectives that need to be attained are in the proposed system as follows,

- [1] EEG Signal Acquisition Module - Add an EEG sensor module into the smart helmet to record brainwaves throughout the rides.
- [2] Real-time Wireless Data Transmission - Introduce Bluetooth low energy (BLE) communications to send EEG data in a stable and efficient manner to the companion mobile application.
- [3] Stress Classification Model - Build and Train a Machine learning model that will be able to analyze EEG pattern, differentiate between stress and motion artifact, and accurately determine stress levels in real-time.
- [4] Alert and Intervention Mechanism - Develop feedback mechanisms to send immediate alert and interventions in form of guided breathing, rest reminders, hydration reminders, or soothing sounds in case of high levels of stress.
- [5] Mobile Application Interface - This will present the mobile application with an easy-to-use interface that presents a color-coded metric with real-time current stress-level values to generate awareness and awareness to the rider in the short-term and long-term to make sure relieved and not stressed.
- [6] System Robustness and Personalization - Provide adaptive learning and error handling to support the variability of individuals in stress patterns, support low false positives/negatives, and the stability of performance in the real-world riding conditions.

3.2 System Overview Diagram

This project involves the smart helmet system intelligent system that provides safety and well-being of the rider since it combines real-time health status, emotional state, danger zones, and post-ride risk analysis. In contrast to the traditional online helmet, the proposed solution constantly analyzes the physical and mental condition of the rider with affordable, wearable biosensors and anticipates accidents and cumulative health risks with proactive feedback to avoid them.

Such sensors as heart rate, body temperature, EEG (stress detection) and IMU (accelerometer, gyroscope, magnetometer) are integrated into the system to capture health, emotional load, and riding habits. An ESP32 microcontroller incorporates edge computing, which reads data of the sensors and sends it to a smartphone app using the Bluetooth Low Energy (BLE) communication standard. Hands-free emergency commands such as the ability to say, "Send Help" or "Call Contact" when help is needed, in health life-and-death situations is possible with voice recognition.

The helmet can pre-recognize dangerous areas based on previous route information and display them in the application by spatial indexing (R-tree/KD-tree) before riding. In real time, it is tracked by GPS as well as use the text-to-speech voice alerts when the rider is in a risk area in the language of choice (English or Sinhala) during rides. Various relaxation suggestions and feedback are given in an emotionally adaptive manner as per the analysis of the EEG.

The information recording journeys along with added external weather and traffic API data after each ride are combined into a risk report that gives a personal risk assessment on unsafe segments, stress peaks, and behavioral patterns. The use of weekly summaries allows detecting chronic health or stress issues in an early stage. It is an intelligent system which turns the helmet into an active health companion and extends the gap between personal mobility and active healthcare monitoring. It is based on the world data on road safety (WHO) and is confirmed by the studies in IoT-based health monitoring (IEEE).

3.3 Individual Component

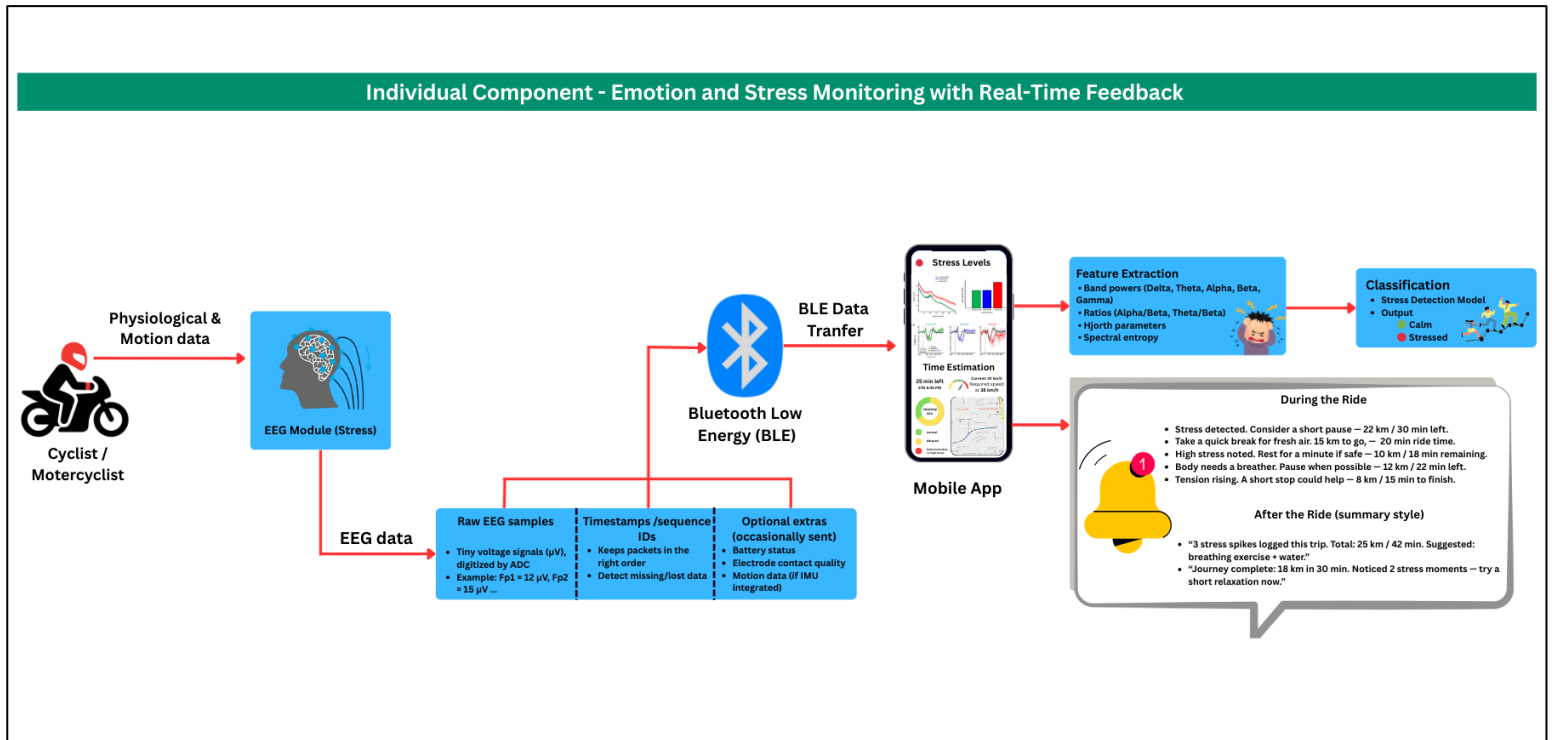


Figure 3: Individual Component Diagram

3.3.1 The flow of the system diagram – Individual Component

The systematic approach to the implementation of EEG devices into one of the smart helmets with BLE connectivity and a mobile app will allow detecting the elevated stress levels of the rider in the most accurate ways and provide prompt intervention capabilities, following the routinized organizational norms.

The first involves having the EEG electrodes hidden in the inner lining of the smart helmet as constant monitors of the brainwave activity of rider in real time during the ride. These electrodes record electrical data related to the cognitive and emotional conditions and send to an onboard microcontroller to be preprocessed.

When there is an incoming stream of raw EEGs, the microcontroller runs noise suppression and artifact elimination algorithms to get rid of deformations brought about by movement, vibrations, or any potential interferences of the environment. These signals after being cleaned are packaged and sent in real time using Bluetooth Low Energy (BLE) to the companion mobile app.

After being picked up by the mobile application, EEG data are fed into a stress classification model based on machine learning. It is a model to study the brainwaves, compare it to datasets trained models, and identify accurately how stressed a rider is at any time. According to this classification, the system measures the nature of the stress situation to determine whether it is necessary to intervene in:

The mobile application immediately brings interventions on board when high levels of stress are realized. These can be discreet warning, breathing prompts, drinking prompt, posture tip or refueling the tank prompt. The application also visualizes the stress levels in real-time using a color-coded interface thus keeping the rider constantly aware of his/her cognitive awareness.

This disciplined approach can be used to integrate their chosen system of EEG electrodes, an onboard microcontroller, Bluetooth Low Energy-based communication, and mobile application, which all act in synchronization and produce real-time monitoring and individualized interventions. This active setup turns the helmet not just into a passive accessory to enhanced safety but also into a smart system that most likely will enhance awareness of riders, reduce stress-related risks, and drive towards safer and healthier motorcycle rides.

3.4 Tools and Technologies

Table 2: Tools and Technologies

Tools	Technologies	Explanation
Arduino Boards 		The Arduino board stands as a compact yet potent microcontroller platform, celebrated for its straightforwardness and adaptability. Boasting a microcontroller unit (MCU) alongside an array of input/output pins, it provides users with effortless access to interfacing with a diverse range of sensors and actuators. With its intuitive development environment and extensive repository of open-source code, even novices can swiftly grasp programming concepts and embark on project development. Available in various configurations and specifications, the Arduino board caters to a broad spectrum of project needs. Bolstered by an engaged community of users and resources, it retains its status as a favored choice for electronics prototyping and DIY endeavors worldwide.
EEG Sensor 		EEG sensor is one of the most important tools used to record the electrical activity of the brain, which makes it possible to monitor changes in the cognition and emotional status of a person in real-time. As part of the smart helmet systems, these sensors are instrumental in identifying the mental and emotional stress levels of the riders by identifying brainwave patterns, which are alpha, beta, theta, and delta rhythms. EEG sensors are strategically integrated into the helmet lining and continuously record neural activity throughout the riding process without any uncomfortable or distractive effect. Such proactive sensory enables the detection of undue stress and fatigue, and hence early preventative measures can be put in place to ensure rider safety. EEG sensors can turn a traditional passive protective safety device (a helmet) into an active safety protection device (both of safety

		of the mind and bodily safety) since by translating complex sensor data into simple solutions and insights in the area of brain activity that can easily be turned into actionable tasks that a helmet can perform to make the use of a helmet is especially relevant in this case since the helmet itself is then being used to guard not just against physical danger but also against cognitive danger.
Firebase 		Firebase is commonly known to be one of the building blocks of mobile and web platform development since it is user-friendly and allows data to update in real-time. Firebase is a platform that allows developers to develop, enhance and grow apps with absolute ease and it is developed and put up by Google. When doing work with authenticating the user, working with cloud databases, hosting content on the Internet, or sending push notifications, the Firebase offers a unified platform in which apps are developed with high efficiency. Its real-time database and Fire store also make it easy to synchronize data between devices in real-time, and there is also integrated data analytics and crash reporting. The declarative APIs and cross-platform capabilities of the platform make it evaluable to developers of all levels of proficiency, easing the difficulty of backend work. Supporting this is a strong body of documentation, widespread community support and deep connection to the rest of Google Cloud, making Firebase a low hanging fruit of modern application development to achieve scalable, secure, and user-friendly digital experiences.
	Bluetooth Low Energy (BLE) 	Android, a ubiquitous operating system developed by Google, represents a cornerstone in the realm of mobile technology, esteemed for its ubiquity and versatility. Renowned for its intuitive user interface and expansive ecosystem of applications, Android powers a vast array of smartphones, tablets, and other mobile devices worldwide. With its open-source nature and robust development framework, Android offers developers the flexibility to create innovative

		and feature-rich mobile applications tailored to diverse user needs. From productivity tools to entertainment apps, Android provides a platform for seamless integration and interaction, facilitating a seamless user experience across various devices and form factors. Supported by a vibrant community of developers and resources, Android continues to shape the landscape of mobile computing, driving innovation and connectivity on a global scale.
Arduino IDE 		Arduino Integrated Development Environment (IDE) is one of the most important tools used in the sphere of electronics prototyping because of its user-friendly interface and efficiency. Being the first software to create programming Arduino microcontrollers, the IDE does not disappoint since it offers simple interface consisting of a package of powerful features in use to the end users, and a system of features customized according to the needs of the novice and the experienced developers. The Arduino IDE painlessly accomplishes the tasks of writing, compiling and uploading code to Arduino boards with features including an intuitive code editor, syntax highlighting, and an integrated serial monitor. Its large database of libraries and example codes written before also speed up the development process such that the user can easily utilize in their projects the complex functionalities that are prewritten in the libraries. With user-friendly interface, a strong community of users, and comprehensive documentation, the Arduino IDE software has become a standard tool in the toolbox of hobbyists, designers, and professionals alike: it makes it easy and efficient to turn creative visions into reality!
VS Code 		Visual Studio Code (VS Code) is a lightweight integrated development environment (IDE) created by Microsoft and known to be very versatile and cross-platform friendly. Having been made such that it can serve the varied development requirements, VS Code fits a broad range of programming language and

		frameworks, and hence its high flexibility in web application development, mobile application development, and cloud application development. The major functions of it are the possibility of intelligent code completion IntelliSense, in-built Git, and powerful debugger, which eases writing the code and sharing. One of the brightest advantages of VS Code is its extension marketplace, in which a developer can add specialized Flutter, Dart, Python, JavaScript, and many other toolkits to it. The availability and accessibility through Windows, macOS and Linux, actively supported community, and frequent updates will allow the cross-platform compatibility and provide constant enhancement. With its simplicity coupled with flexibility, VS Code has emerged as one of the most popular development platforms that enable developers to be creative and to be innovative within the dynamic changing software environment in a very effective way.
Flutter  Flutter		Flutter, another development with open-source framework, one of the earliest frameworks created by Google, flutter has rapidly become a pillar of contemporary application development. Famous due to its capability of developing cross-platform applications out of one codebase, Flutter helps software engineers develop frictionless, working well on applications on Android, iOS, web and on desktop platforms. It is built on the Dart programming language with a rich set of pre-designed widgets producing expressive and flexible user interfaces. It has a very high level of the development cycle and as such offers hyper-fast Hot Reload, enabling developers to see the change at once, hence improving productivity and creativity. It is architected on high fidelity of application performance and having native user experience that apps are easy to transfer between locations and platforms. Google is sponsored and maintained by an active community of developers around the world; Flutter is PACKED with documentation and never-ending

		updates and a HUGE ecosystem of packages. Combining efficiency, adaptability, and innovation, Flutter keeps redefining cross-platform development and empowering developers to present powerful solutions within record timeframes and with a high degree of accuracy.
Jupyter Notebook 		Jupyter Notebook is a well-recognized open-source interactivity framework, and is hailed as universal across data science, machine learning, and scientific computing tasks. It gives a simple interface, which can be used by developers and researchers to mingle live code, narrative text, visualizations, and equations in the same document. This combination of codes and documentation into one environment supports exploration as well as reproducibility encouraging and promoting Jupyter Notebook as one of the essential tools in research and analysis. Especially useful with signal treatment and machine learning classification, Jupyter Notebook makes it possible to evaluate EEG data using libraries like NumPy, SciPy, TensorFlow and Scikit-learn. It has a cell-based workflow, which permits users to experiment with algorithms, visualize EEG patterns, and refine classification models in an iterative manner with real-time feedback. The fact that the platform supports several programming languages, and above all, Python, is flexible on various projects. Now with a lot of extensions and active community support, Jupyter Notebook can still give developers and researchers a user-friendly, collaborative, and effective environment to further data-driven research and system development.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

[1]. EEG Signal collection and Transmission

- The system should be able to monitor brainwaves in EEG sensors permanently embedded into the helmet.
- It must transmit preprocessed EEG data to the mobile application by use of Bluetooth Low Energy (BLE).
- Real-time data transmission is desirable with latency time as minimal as possible to facilitate timely stress detection.

[2]. Stress Detection Classification

- The system must be able to interpret EEG patterns based on trained machine learning model
- It ought to be precise in distinguishing between normal, elevated and high-stress conditions.
- Customization of stress type must be according to the rider profile, to achieve accuracy in detecting the stress type.

[3]. Composability with components

- The system must connect and hopefully merge EEG sensors, a microcontroller, BLE communication and the application on the phone.
- It must guarantee ease in data flow and synchronization of hardware and software components.

[4]. Alert Mechanism

- When the level of stress is high, the system must issue warnings by way of discreet vibration or sounds.
- It should be able to deliver customized interventions, e.g. guided breathing cues, hydration reminders or safe-stop suggestions.
- It is required that the alerts switch off automatically when the rider becomes stress-free.

[5]. Database Management

- The application will have to use a secure database, as part of the mobile app, to store stress data after each ride.
- It must capture timestamps, stress levels and contextual information that can be analyzed and reviewed in future.
- This data should be then displayed in the mobile app with the use of easy-to-understand color-coded dashboards and post-ride summaries.

4.1 Non- Functional Requirements

[1]. Performance

- The system is required to process EEG signals and produce alerts in real time, and the end-to-end latency should not be more than 300 ms.
- Notifications and interventions need to be sent immediately after the detection of stress and help them to be aware of it in time.

[2]. Reliability

- The system is required to be used under the same conditions consistently in terms of motion, vibration, sweat and different environmental conditions (0 to 45 degrees Celsius).
- Classification of stress should show high accuracy by reducing false positive or false negative scenarios which mislead the rider.

[3]. Scalability

- The system must be open and capable of incorporating other biosensors (e.g. heart rate, IMU) without degrading performance.
- The loaded mobile application will have to manage several rider accounts, and it will have to work with historical data.

[4]. Security

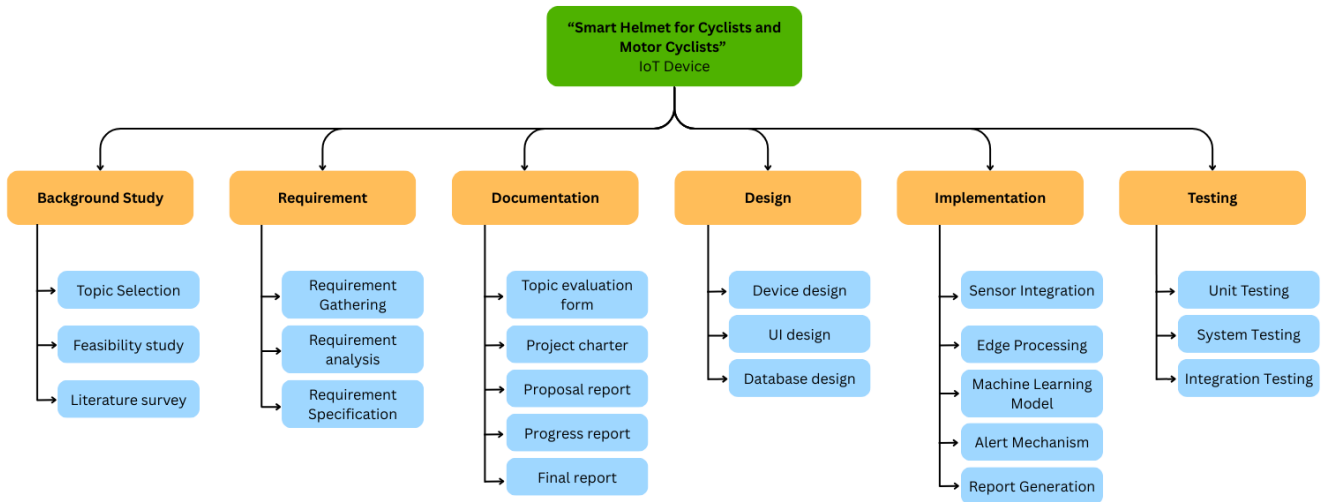
- EEG data sent using BLE will have to be encrypted (e.g., by AES-128/256) to avoid any unauthorized access or modification.
- Providing users and system configuration access to non-authenticated users must be limited.

[5]. Usability

- Alerts should be discreet and free of distraction and must be presented in the form of haptic or short audio alerts.
- It is recommended that the mobile application interface should have simple stress indicators (colors) which should be intuitive to most first-time users and should be simple to set up.

5. WORK BREAKDOWN CHART

Figure 4: Work breakdown chart



6. GANTT CHART

Table 3: Gantt Chart

No	Task Name	Start Date	End Date		2025-2026													
				May	Jun	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
1	Topic evaluation	10-May	23-July															
1a	Select a topic	10-May	26-May															
1b	Select a supervisor	20-May	26-May															
1c	TAF Submission	2-Jun	27-Jun															
1d	Project charter submission	29-June	23-July															
2	Project proposal report	20-24 July	8-12 Sep															
2a	Create Project Proposal - Group	20-24 July	20-23 Aug															
2b	Create Project Proposal - individual	20-24 July	20-23 Aug															
2c	Project proposal presentation	8- Sep	12- Sep															
3	Develop the system	15-20 Sep	9-17 Mar															
3a	Identifying functions	15-20 Sep	12-25 Sep															
3b	Database Designing	15-20 Sep	1-5 Oct															
3c	Implementation	1-5 Oct	25-30 Jan															
3d	Unit testing	1- 5 Feb	26- 28 Feb															
3e	Integration testing	1- 5 Aug	25- 30 Mar															
4	Progress Presentation - I	3- 17 Oct	3- 17 Jan															
4a	Project Status document	3- 17 Oct	25- 28 Dec															
4b	Create presentation document	3- 17 Oct	25- 28 Dec															
4c	Progress Presentation – I (50%)	3-Jan	17-Jan															
5	Research Paper	1- 5 Sep	10- Dec															
5a	Create the Research Paper	1- 5 Sep	10- Dec															
6	Progress Presentation - II	10- 16 Feb	2- 16 May															
6a	Create presentation document	10- 16 Feb	27- 30 Apr															
6b	Progress presentation – II (90%)	10- 16 Mar	2- 16 May															
7	Final Report Submission	16-May	11-july															
7a	Create final presentation	16-May	20-June															
7b	Final Report Submission	16-May	20-June															
7c	Application assessment	16-May	20-June															
7d	Project status document	16-May	20-June															
7e	Student logbook	16-May	20-June															
7f	Final Presentation & Viva	20-June	11-july															

7. BUDGET AND BUDGET JUSTIFICATION

The budget of the development and implementation process of EEG-based Smart Helmet is multifaceted because it consists of numerous aspects which should be outlined, such as project scope, technology aspects, personnel, infrastructure, and operational cost. It is not possible to define an exact budget without elaborate technical and financial specifications, but one can describe some important areas of costs. These are EEG sensor modules, microcontroller units, mobile application development, machine learning model training, testing and evaluation costs, maintenance and possible cloud-based data storage solutions.

Component	Amount
Body & Ambient Temperature Sensors	Rs.370
Pulse Sensor	Rs.350
EEG Module	Rs.27000
IMU Sensor	Rs.1500
Microcontroller (ESP32)	Rs.760
Firebase	\$2/Month – Rs. 650 (Pay as you go)
Helmet	Rs.6000
Arduino Uno board	Rs.2550
Total	Rs.39,480

Table 4: Budget and Budget Justification

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9. APPENDICES

feedback studio
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**SST BASED SMART HELMET FOR
CYCLISTS AND MOTOR CYCLISTS: A CASE STUDY**

25-26J-294

Project Proposal Report

Samarakoon S S A D S R

Match Overview

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